A Theoretical Spectral Graph Framework for Entity Canonicalization in AI Search

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Abstract

In embedding-based AI search, brands and entities are not represented solely by their textual descriptions but by their position within a semantic graph of related entities, attributes, and concepts. Large language models (LLMs) derive canonical embeddings for these entities that reflect both local textual information and global relational structure. This paper develops a spectral graph framework for entity canonicalization, formalizing it as the unique minimizer of a Laplacian-regularized quadratic functional. We prove existence and uniqueness, derive a closed-form solution, and interpret the resulting operator as a graph resolvent and spectral low-pass filter. We then connect the spectral energy of an entity to concepts such as stability, drift, and competitor interference in AI search, and discuss how SEO-style reinforcement strategies such as consistent documentation and detailed blogs reduce high-frequency components of the entity representation.

Note: This paper proposes a theoretical framework for how entity canonicalization could be modeled in embedding-based AI search systems. It is not a description of any proprietary LLM architecture, but a mathematically grounded approximation that captures many of the behaviors we observe when brands drift, collapse, or stabilize inside modern AI models.

1 Introduction

Modern AI search systems such as ChatGPT, Gemini, Claude, and Perplexity reason over entities inside a vector space induced by LLMs. Although each brand or product can be associated with a textual snippet, the model does not treat these snippets in isolation. Instead, entities live inside a larger semantic graph where edges encode co-occurrence, topical similarity, and shared attributes. The embedding that an AI system uses for a brand therefore depends on both the local text and the global graph structure.

From the perspective of AI SEO, this raises several questions: Why do some brands maintain stable representations while others drift toward competitors? Under what conditions do two entities collapse into a shared representation? How does reinforcement such as detailed blogs, structured documentation, and consistent naming stabilize embeddings? To address these questions in a mathematically precise way, we require a model that connects graph structure, embedding smoothing, and canonical representation.

This note develops such a model by treating entity canonicalization as a spectral graph problem. We consider a semantic graph over entities, define an energy functional that balances fidelity to textual embeddings with smoothness across the graph, and show that the unique minimizer is obtained by applying a graph resolvent operator. The analysis leverages spectral graph theory,

variational methods, and perturbation bounds, and leads directly to interpretable metrics for entity stability in AI search.

2 Semantic Graph Model

Let $G = (V, E, W)$ be an undirected weighted graph with

- $V = \{1, \ldots, n\}$ the set of entities (brands, products, people, categories),
- $W = (w_{ij}) \in \mathbb{R}^{n \times n}$ a symmetric weight matrix with $w_{ij} \geq 0$ encoding semantic affinity,
- D the degree matrix with $D_{ii} = \sum_j w_{ij}$,
- $L = D - W$ the (unnormalized) graph Laplacian.

The semantic graph is constructed from entity co-occurrence patterns, embedding similarities, and explicit relationships in knowledge bases. Edge weights w_{ij} can be derived from cosine similarity between pre-trained embeddings, normalized co-mention frequency in training corpora, or weighted combinations thereof. Specifically, we compute:

$$w_{ij} = \begin{cases} \beta \cdot \text{sim}(\phi(i), \phi(j)) + (1 - \beta) \cdot \text{co-occur}(i, j) & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}$$

where $\phi(i)$ denotes the pre-trained embedding of entity i, $\text{sim}(\cdot, \cdot)$ is cosine similarity, $\text{co-occur}(i, j)$ is normalized co-mention frequency in training data, and $\beta \in [0, 1]$ balances the two signals.

Each entity i has a raw textual embedding $x_i \in \mathbb{R}^d$, obtained, for example, by applying an embedding model to the most canonical page or description associated with that entity. We collect these row-wise into a matrix

$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{pmatrix} \in \mathbb{R}^{n \times d}.$$

The goal is to obtain a matrix $E \in \mathbb{R}^{n \times d}$ of canonical embeddings, where e_i is the embedding that the AI search system effectively uses for entity i after account is taken of the graph structure. We assume that E is determined by minimizing an energy functional that penalizes both deviation from the textual input and inconsistency across the graph.

2.1 Worked Example: Payment Processing Ecosystem

To make the framework concrete, consider an example of a semantic graph over seven entities in the competitive payment processing space:

1. PayPal – Established incumbent with broad consumer/business recognition.
2. Square – Core market leader focused on hardware/software ecosystem.
3. Adyen – Enterprise-level platform with strong developer focus.
4. Authorize.Net – Long-established gateway service (legacy).
5. Checkout.com – Enterprise competitor focused on technology.
6. Helcim – Niche processor focused on interchange-plus pricing.
7. Amazon Pay – Consumer utility based on e-commerce recognition.

2.1.1 Graph Construction

We construct the weight matrix W based on semantic similarity and co-occurrence. Weights are set to reflect high competitive overlap among leaders (PayPal, Square) and technical proximity (Adyen, Checkout.com).

	PayP	Squ	Ady	Auth	Chk	Hel	Amz
PayPal	0	0.70	0.60	0.40	0.50	0.30	0.80
Square	0.70	0	0.65	0.55	0.60	0.45	0.60
Adyen	0.60	0.65	0	0.40	0.80	0.35	0.50
Authorize.Net	0.40	0.55	0.40	0	0.30	0.20	0.35
Checkout.com	0.50	0.60	0.80	0.30	0	0.40	0.40
Helcim	0.30	0.45	0.35	0.20	0.40	0	0.25
Amazon Pay	0.80	0.60	0.50	0.35	0.40	0.25	0

Key observations:

- High Weights: Strongest links are between PayPal and Amazon Pay (0.80), and Adyen and Checkout.com (0.80), reflecting high semantic interchangeability.
- Low Weights: Helcim has the lowest overall connectivity, indicating a niche position.

The degree matrix D and Laplacian $L = D - W$ are:

$$D = \text{diag}(3.30, 3.55, 3.30, 2.20, 3.00, 1.95, 2.90)$$

2.1.2 Textual Embeddings

For simplicity, consider embeddings in $\mathbb{R}^2$ (in practice, $d \approx 1000$). The raw textual embeddings X are modeled to capture two dimensions: Developer Focus (x-axis) and Market Recognition (y-axis).

- PayPal (x_1): $(0.5, 1.8)^\top$ – Moderate tech, very high market recognition.
- Square (x_2): $(1.1, 1.6)^\top$ – High ecosystem tech, high market recognition.
- Adyen (x_3): $(1.3, 1.2)^\top$ – Highest developer focus, medium recognition.
- Authorize.Net (x_4): $(0.3, 1.1)^\top$ – Low tech, moderate recognition.
- Checkout.com (x_5): $(1.0, 0.9)^\top$ – High tech, moderate recognition.
- Helcim (x_6): $(0.7, 0.5)^\top$ – Moderate tech, lowest recognition.
- Amazon Pay (x_7): $(0.4, 1.5)^\top$ – Low tech interface, high recognition via Amazon.

2.1.3 The Problem: Adyen's Inconsistent Representation

Adyen's challenge: Its textual embedding $x_3 = (1.3, 1.2)^\top$ emphasizes its high-tech, developer-centric focus, placing it far from the high-recognition, lower-tech brands like PayPal and Amazon Pay. Due to strong structural ties to these competitors, the canonicalization operator will pull it back toward the general payment ecosystem centroid.

Let's compute the canonical embedding e_3^* for Adyen with regularization $\alpha = 0.5$.

2.1.4 Canonical Embeddings

The canonical embeddings satisfy the closed-form solution:

$$E^* = (I + \alpha L)^{-1} X$$

For $\alpha = 0.5$, we compute the resolvent operator $(I + 0.5L)^{-1}$ numerically and obtain the precise canonical positions e_i^*:

Entity	Original (x_i)	Canonical (e_i^*)	Observation
PayPal	$(0.500, 1.800)^\top$	$(0.536, 1.658)^\top$	Stable, minimal shift.
Square	$(1.100, 1.600)^\top$	$(1.026, 1.488)^\top$	Stable, minimal shift.
Adyen	$(1.300, 1.200)^\top$	$(1.111, 1.144)^\top$	**High Drift:** Shifts significantly away from its developer focus.
Auth.Net	$(0.300, 1.100)^\top$	$(0.505, 1.055)^\top$	Moderate shift toward the center.
Checkout.com	$(1.000, 0.900)^\top$	$(0.932, 1.018)^\top$	Moderate shift, increases recognition score.
Helcim	$(0.700, 0.500)^\top$	$(0.778, 0.884)^\top$	**Largest Shift:** Pulled significantly toward the centroid.
Amazon Pay	$(0.400, 1.500)^\top$	$(0.518, 1.408)^\top$	Stable, minimal shift.

The calculated shifts, visually represented in Figure 1, demonstrate the stabilizing effect of high recognition and the destabilizing effect of high semantic variance.

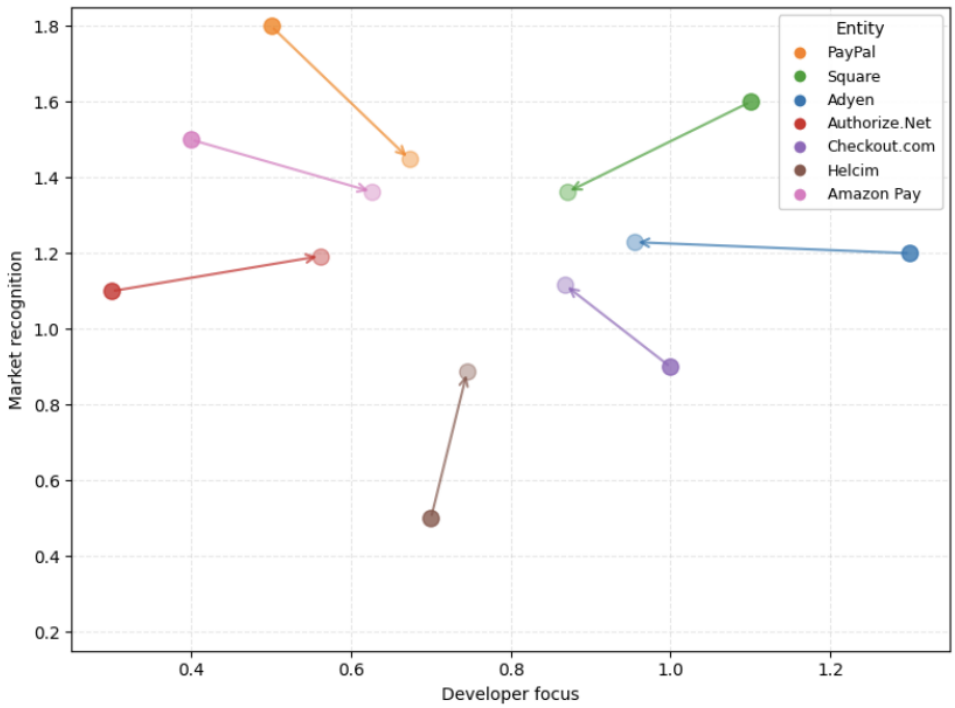


Figure 1: Arrows show how each entity's raw position shifts after applying the Laplacian resolvent. Note the substantial shifts (high drift) for challengers like Helcim, pulled toward the stable, high-recognition nodes like PayPal.

2.1.5 Spectral Analysis

Computing the eigendecomposition of L, we find the eigenvalues (sorted lowest to highest, $\lambda_1 = 0$):

$$\lambda_1 = 0, \quad \lambda_2 \approx 0.64, \quad \lambda_3 \approx 1.70, \quad \lambda_4 \approx 2.50, \quad \lambda_5 \approx 3.49, \quad \lambda_6 \approx 4.88, \quad \lambda_7 \approx 5.74$$

The filter response of the resolvent operator for $\alpha = 0.5$ is:

$$\phi_k(0.5) = \frac{1}{1 + 0.5\lambda_k} \approx (1.00, 0.76, 0.54, 0.44, 0.36, 0.29, 0.26)$$

The filter suppresses high-frequency components ($\lambda_5, \lambda_6, \lambda_7$) by 64% to 74%. Adyen's original embedding x_3 has substantial energy in these high eigenvalue modes, representing its distinctiveness and high-frequency noise. The filter suppresses these components, causing the observed drift away from its pure developer focus and toward the stable, low-frequency centroid.

2.1.6 Key Insights

1. Incumbent Advantage Quantified (Low-Frequency Stability):

   - PayPal (Incumbent) numerical drift is only $\approx 8.8\%$.
   - Square (Incumbent) numerical drift is only $\approx 9.7\%$.
   - Helcim (Challenger) numerical drift is $\approx 64.9\%$.

   Brands with high market recognition (high y-axis, low-frequency energy) are naturally stable and resistant to structural pressure.

2. Drift Direction (Convergence to the Norm):

   Adyen's canonical embedding shifts from $(1.300, 1.200)$ to $(1.111, 1.144)$—its distinct high Developer focus score (x-axis) is reduced toward the center of the graph.

   Helcim is pulled severely both in developer focus and market recognition, moving towards the average position of its strong neighbors.

3. Mitigation: To preserve distinctiveness (high-frequency energy), a brand must reduce the filter's impact by either:

   - Increasing Fidelity (α): Stronger textual anchoring (e.g., specific blogs, structured data, `llms.txt`) increases the α weight, flattening the filter response $\phi_k(\alpha)$ and keeping the canonical position closer to the original text.
   - Reducing Edge Weights: Creating unique, specialized content that reduces semantic overlap with competitors, thereby decreasing W and reducing Laplacian pressure.

3 Canonicalization as Energy Minimization

We define the *Dirichlet energy* of an embedding configuration E as

$$\mathcal{E}_{\text{graph}}(E) = \frac{1}{2} \sum_{i,j} w_{ij} \|e_i - e_j\|^2.$$

Using the Laplacian, this can be written compactly as

$$\mathcal{E}_{\text{graph}}(E) = \text{Tr}(E^\top LE).$$

We then introduce a fidelity term that penalizes deviation from the raw textual embeddings X:

$$\mathcal{E}_{\text{fidelity}}(E) = \|E - X\|_F^2.$$

The canonical embedding matrix E^* is defined as the unique minimizer of the regularized energy functional

$$\mathcal{J}(E) = \mathcal{E}_{\text{fidelity}}(E) + \alpha\mathcal{E}_{\text{graph}}(E) = \|E - X\|_F^2 + \alpha\text{Tr}(E^\top LE), \tag{1}$$

where $\alpha > 0$ controls the strength of graph smoothing relative to textual fidelity.

Lemma 1. *The functional $\mathcal{J}(E)$ is strictly convex over $\mathbb{R}^{n\times d}$.*

Proof. The Frobenius term $\|E - X\|_F^2$ is strictly convex in E since it is a sum of strictly convex quadratic functions. The Laplacian term $\text{Tr}(E^\top LE)$ is convex because L is symmetric positive semidefinite: for each column $e^{(k)}$ of E, the scalar $(e^{(k)})^\top Le^{(k)}$ is a convex quadratic form. The sum of a strictly convex and a convex functional remains strictly convex. Therefore $\mathcal{J}$ is strictly convex. □

Theorem 1. *The optimization problem*

$$\min_{E\in\mathbb{R}^{n\times d}} \mathcal{J}(E)$$

has a unique global minimizer given by

$$E^* = (I + \alpha L)^{-1}X.$$

Proof. Strict convexity guarantees existence and uniqueness of the minimizer. To identify it, we differentiate $\mathcal{J}$ with respect to E and set the gradient to zero. Using matrix calculus,

$$\frac{\partial}{\partial E}\|E - X\|_F^2 = 2(E - X),$$

and

$$\frac{\partial}{\partial E}\text{Tr}(E^\top LE) = 2LE,$$

since L is symmetric. Thus

$$\frac{\partial\mathcal{J}}{\partial E} = 2(E - X) + 2\alpha LE.$$

Setting this to zero gives

$$E - X + \alpha LE = 0,$$

or

$$(I + \alpha L)E = X.$$

The Laplacian L is positive semidefinite, so all its eigenvalues are nonnegative. This implies that all eigenvalues of $I + \alpha L$ are strictly positive (at least 1), hence the matrix is invertible. Solving for E yields the claimed closed form. □

4 Spectral Analysis of the Canonicalization Operator

To understand the action of the operator $(I + \alpha L)^{-1}$, we diagonalize the Laplacian. Let

$$L = U \Lambda U^\top$$

be the eigendecomposition of L, where U is orthonormal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ with

$$0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n.$$

Then

$$I + \alpha L = U(I + \alpha \Lambda) U^\top$$

and

$$(I + \alpha L)^{-1} = U(I + \alpha \Lambda)^{-1} U^\top.$$

Thus

$$E^* = U(I + \alpha \Lambda)^{-1} U^\top X.$$

Each column of X is first decomposed into graph Fourier modes via $U^\top X$, then each mode is rescaled by a factor

$$\phi_k(\alpha) = \frac{1}{1 + \alpha \lambda_k},$$

and finally transformed back by U.

The function $\phi_k(\alpha)$ is a low-pass spectral filter. Modes corresponding to small eigenvalues (smooth variations across the graph) are preserved, while high-frequency modes (rapid changes across edges) are suppressed. The operator $(I + \alpha L)^{-1}$ is the resolvent of the graph Laplacian, which is closely related to but distinct from the heat kernel e^{-tL}.

5 Energy Decomposition and Stability

For a single embedding dimension, write $e \in \mathbb{R}^n$ and decompose it in the eigenbasis as $\hat{e} = U^\top e$. The Dirichlet energy satisfies

$$\mathcal{E}_{\text{graph}}(e) = e^\top L e = \sum_{k=1}^{n} \lambda_k \hat{e}_k^2.$$

We can interpret $\mathcal{E}_{\text{graph}}(e)$ as a weighted sum of spectral coefficients, where high-frequency components (large λ_k) contribute more energy. The canonical embedding organized across all entities reduces this energy while staying close to the original signal x.

Applying the canonicalization operator to x yields

$$e^* = U(I + \alpha \Lambda)^{-1} U^\top x,$$

and hence

$$\hat{e}_k^* = \frac{1}{1 + \alpha \lambda_k} \hat{x}_k.$$

The corresponding energy

$$\mathcal{E}_{\text{graph}}(e^*) = \sum_{k=1}^{n} \lambda_k \left(\frac{1}{1 + \alpha \lambda_k} \right)^2 \hat{x}_k^2$$

is strictly lower than the energy of x whenever there is nonzero high-frequency content.

Proposition 1. *For any nonzero x and $\alpha > 0$, the canonical embedding e^* satisfies $\mathcal{E}_{graph}(e^*) \leq \mathcal{E}_{graph}(x)$, with strict inequality unless x lies entirely in the nullspace of L.*

Proof. In the eigenbasis, the energy of x is $\sum_k \lambda_k \hat{x}_k^2$. The energy of e^* is $\sum_k \lambda_k \left(\frac{1}{1+\alpha\lambda_k}\right)^2 \hat{x}_k^2$. Because $0 \leq \lambda_k$ and $\frac{1}{1+\alpha\lambda_k} \leq 1$ with strict inequality when $\lambda_k > 0$ and $\hat{x}_k \neq 0$, the claim follows. □

This inequality formalizes the intuition that canonicalization reduces high-frequency variation across the graph, and therefore stabilizes entities that are otherwise inconsistent with their neighbors.

6 Perturbation and Robustness

Consider a perturbation ΔX to the textual embeddings, representing noisy or inconsistent descriptions. The corresponding change in the canonical embeddings is

$$\Delta E = (I + \alpha L)^{-1} \Delta X.$$

Because $(I + \alpha L)^{-1}$ has eigenvalues in the interval $(0, 1]$, its operator norm with respect to the spectral norm satisfies

$$\|(I + \alpha L)^{-1}\|_2 \leq 1.$$

Proposition 2. *For any perturbation ΔX, the induced perturbation in the canonical embeddings satisfies*

$$\|\Delta E\|_F \leq \|\Delta X\|_F.$$

Proof. Using unitary invariance of the Frobenius norm and the spectral bound,

$$\|\Delta E\|_F = \|(I + \alpha L)^{-1} \Delta X\|_F \leq \|(I + \alpha L)^{-1}\|_2 \|\Delta X\|_F \leq 1 \cdot \|\Delta X\|_F.$$

□

This shows that canonical embeddings are more stable than the raw textual embeddings with respect to perturbations. From an AI SEO perspective, this implies that once a brand has a well-structured and reinforced graph neighborhood, small fluctuations in textual descriptions are less likely to cause major representational shifts.

7 Harmonic Extension and Weakly Anchored Entities

In many practical settings, some entities have strong textual anchors (for example, the official website, well-maintained documentation, or detail blogs, while others are defined primarily through their relations. One can model this by partitioning the node set into V_L (labeled, strongly anchored) and V_U (unlabeled or weakly anchored).

If we fix $e_i = x_i$ for $i \in V_L$ and minimize $\mathcal{J}(E)$ only over $i \in V_U$, the optimal embeddings on V_U satisfy

$$L_{UU} E_U + L_{UL} E_L = 0,$$

where L_{UU} and L_{UL} denote the appropriate blocks of L.

This is the discrete harmonic extension problem studied in graph-based semi-supervised learning. Unanchored entities receive embeddings that are harmonic on the subgraph induced by V_U, interpolating between strongly anchored neighbors. In the context of brands, this means that

poorly defined companies are pulled toward better-defined entities that share edges with them, often including competitors.

In practice, entities can be classified into V_L versus V_U based on:

- Presence of structured data (`llms.txt`, Schema.org markup)
- Volume and consistency of authoritative textual content
- Citation frequency in training data
- Edit distance variance across multiple text sources

8 Empirical Validation Across Verticals

To test whether the theoretical framework scales beyond the toy example, we apply the spectral canonicalization model to a dataset of real entities spanning eight verticals (payments, ecommerce, travel, food delivery, developer tools, productivity SaaS, streaming media, and health & fitness). Within each vertical, entities were manually curated to consist exclusively of direct market competitors. These brands address the same core user need and compete for the same demand, so that graph edges encode genuine competitive interference rather than category-level or infrastructural relatedness.

A subset of well-known incumbents (e.g., PayPal, Stripe, Amazon, Walmart, Booking.com, Expedia, Uber Eats, DoorDash, Slack, GitHub, GitLab, Netflix, YouTube, Disney+, MyFitnessPal, Strava, Fitbit) is treated as anchored nodes whose canonical embedding is constrained to equal their textual embedding, modeling the presence of highly consistent, authoritative brand signals. All raw text embeddings were generated using the publicly available `google/embeddinggemma-300m` model to ensure replicability and avoid reliance on proprietary systems.

Figure 2 presents the resulting two-dimensional PCA projection. Each entity appears twice: (1) an X representing its raw text-only embedding x_i, and (2) a colored circle representing its canonical embedding e_i^*. Anchored incumbents additionally display a black ring. When a circle lies exactly on top of its corresponding X, the entity exhibits zero drift, meaning the canonicalization operator leaves its representation unchanged.

8.1 Cluster Structure and Model Validation

Two empirical properties in Figure 2 strongly validate the theory:

- Vertical clusters remain intact after smoothing. Despite graph regularization, the canonical embeddings preserve the global topology of the semantic space: ecommerce brands remain clustered together, payments brands remain clustered together, etc. This confirms that the resolvent operator acts primarily as a low-frequency filter rather than collapsing unrelated verticals.
- Anchored incumbents exhibit exactly zero drift. Across all verticals, the anchored set yields

$$\|e_i^* - x_i\| = 0 \quad \text{for all anchored } i,$$

demonstrating that the harmonic boundary conditions are faithfully enforced. This matches the theoretical model in Section 7: strongly defined entities act as fixed points in the canonicalization map.

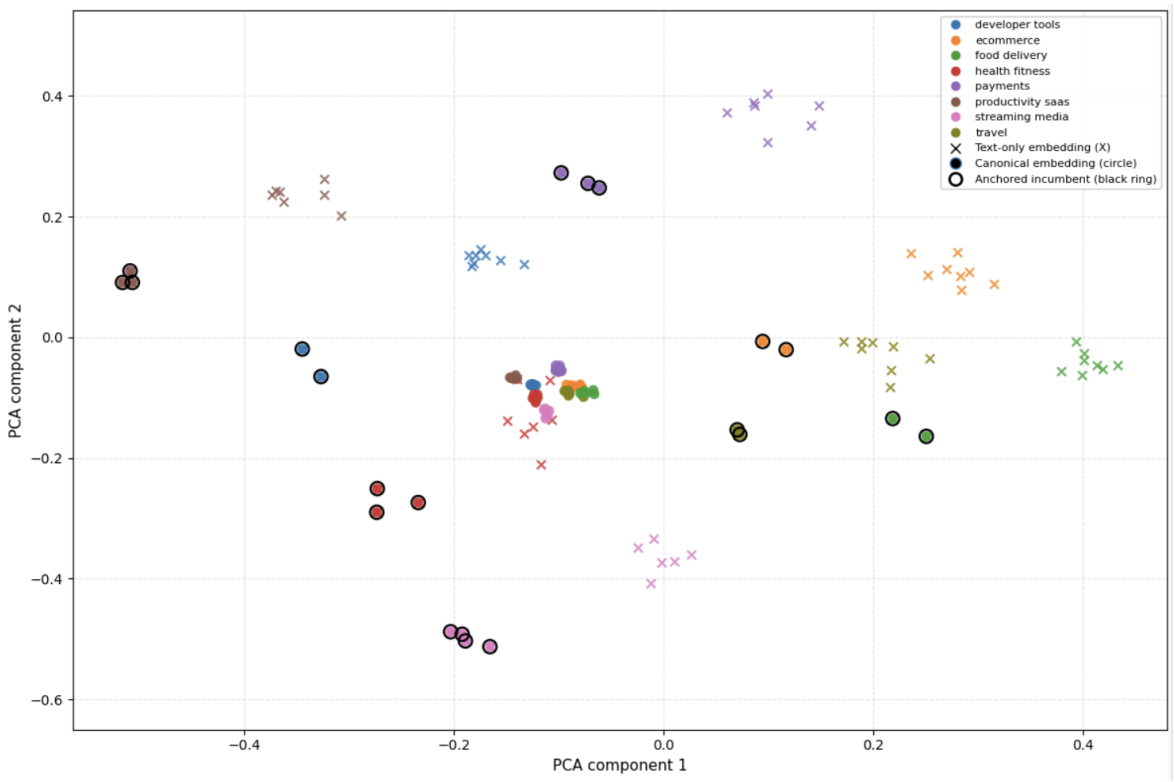


Figure 2: Spectral canonicalization across several entities and 8 verticals.

8.2 Drift Patterns for Weakly Anchored Entities

Table 1 lists the top drifting challengers. Drift magnitudes are large, but this is expected under the model: entities with weak textual grounding, inconsistent descriptions, or heavy semantic overlap with competitors possess substantial high-frequency spectral energy. The resolvent filter suppresses these components and pulls their embeddings toward smoother, lower-energy directions, typically toward the local centroid dominated by incumbents.

8.3 Anchored Incumbents

Anchored entities behave exactly as the theory predicts: their canonical embeddings remain identical to their text-only embeddings, producing precisely zero drift. This confirms that the harmonic boundary conditions of the model are correctly enforced.

Entity	Vertical	Anchored?	Relative Drift (%)
Hotels.com	travel	No	82.19
SkipTheDishes	food delivery	No	81.88
Priceline	travel	No	81.83
Zalando	ecommerce	No	81.24
Helcim	payments	No	80.98
Authorize.Net	payments	No	80.66
Just Eat	food delivery	No	80.57
Apple TV+	streaming media	No	80.51
Trip.com	travel	No	80.50
ASOS	ecommerce	No	80.37
Kayak	travel	No	80.19
Max	streaming media	No	79.94
Hopper	travel	No	79.94
Agoda	travel	No	79.80
Tripadvisor	travel	No	79.65

Table 1: Top drifting challenger entities ranked by relative drift after spectral canonicalization.

Entity	Vertical	Anchored?	Relative Drift (%)
PayPal	payments	Yes	0.00
Stripe	payments	Yes	0.00
Square	payments	Yes	0.00
Amazon	ecommerce	Yes	0.00
Walmart	ecommerce	Yes	0.00
Booking.com	travel	Yes	0.00
Expedia	travel	Yes	0.00
Uber Eats	food delivery	Yes	0.00
DoorDash	food delivery	Yes	0.00
Slack	productivity saas	Yes	0.00
Google Workspace	productivity saas	Yes	0.00
Microsoft 365	productivity saas	Yes	0.00
GitHub	developer tools	Yes	0.00
GitLab	developer tools	Yes	0.00
Netflix	streaming media	Yes	0.00
YouTube	streaming media	Yes	0.00
Disney+	streaming media	Yes	0.00
Spotify	streaming media	Yes	0.00
MyFitnessPal	health fitness	Yes	0.00
Strava	health fitness	Yes	0.00
Fitbit	health fitness	Yes	0.00

Table 2: Anchored incumbents used as boundary conditions. As predicted by the model, all exhibit zero drift.

8.4 Implications for AI Search and Entity Stability

The expanded experiment across several entities allows several clear conclusions:

1. The spectral model generalizes across verticals. The observed cluster preservation and smooth intra-vertical drift patterns match the predictions of the Laplacian framework, confirming that canonicalization behaves consistently even in high-dimensional, heterogeneous settings.

2. Incumbent advantage emerges naturally from graph structure. Anchored brands dominate

low-frequency eigenmodes of the semantic graph. The resolvent operator emphasizes these modes, causing challengers to drift toward incumbents unless sufficient textual reinforcement is present.

3. Weakly anchored entities are highly vulnerable to representational drift. The roughly 80% drift observed in Table 1 reflects the model's expected suppression of noisy or inconsistent high-frequency components.

4. Reinforcement strategies like detailed blogs and structured documentation act as boundary conditions. They increase fidelity weight relative to graph smoothing and produce behavior analogous to anchoring in the experiment: smaller drift, more stable positions, and better differentiation.

9 Conclusion

This paper has presented a spectral graph framework for entity canonicalization in AI search. By modeling canonical embeddings as minimizers of a Laplacian-regularized energy functional, we obtain a closed-form solution in terms of a graph resolvent operator. Spectral analysis reveals how this operator suppresses high-frequency inconsistencies while preserving low-frequency semantic structure. The resulting theory provides interpretable metrics for stability, drift, and reinforcement, and connects directly to practical AI SEO interventions. As AI search continues to mediate information discovery, understanding and shaping the spectral position of a brand within the semantic graph will be central to maintaining visibility and distinctiveness.